PROJECT REPORT: SUPERVISED MACHINE LEARNING & COMMERCIAL INTELLIGENCE

Company: 365 Automobiles Inc. (UAE Showrooms & Service Centres)

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Batch: Batch-06 Capstone Project

1. Executive Summary

365 Automobiles operates four showrooms across Dubai, Abu Dhabi, and Sharjah, supported by three specialized service centres. This project bridges commercial operations and data science by building rigorous supervised machine learning models to predict final deal pricing, identify high-intent sales leads, and classify vehicle model preferences.

Analyzing **12,000 sales enquiries** and **9,178 workshop service records** revealed critical operational insights. Key findings indicate that list prices and customer income strongly anchor final negotiations, digital engagement metrics (test drives and brochure downloads) effectively predict lead conversion, and proactive battery health monitoring is essential for post-sale customer satisfaction.

2. Dataset Overview & Data Preparation

- **Sales Dataset (365_Automobiles_Sales.csv):** Comprises 12,000 records and 39 columns spanning enquiries from January 2024 onward. Features were categorized into identifiers, customer demographics, vehicle specifications, behavioral signals, and outcome fields.
- **Service Dataset (365_Automobiles_Service.csv):** Comprises 9,178 workshop visits across 18 columns, capturing post-sale vehicle maintenance, battery degradation, and service costs.
- **Handling Missing Values & Leakage:** Legitimate missing values (final_price_aed, days_to_delivery, vehicle_id) for non-buyers were strictly preserved and not artificially imputed. To prevent data leakage, feature sets were isolated for each model based strictly on what information is available *at that specific stage* of the customer journey.

3. Machine Learning Models & Results

Model A: Final Selling Price (Regression)

- **Target & Population:** final_price_aed built exclusively on converted buyers ($N = 3,574$).

- **Algorithm:** LinearRegression using pre-negotiation features (e.g., list price, options value, battery specs, customer income, and down payment percentage) while explicitly excluding post-sale and negotiation-tied variables like discounts.
- **Results:** Achieved strong predictive performance with low Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) in AED, effectively capturing vehicle valuation trends across UAE showrooms.

Model B: Lead Conversion (Binary Classification)

- **Target & Population:** purchased (Yes/No) across all 12,000 enquiries.
- **Algorithm:** RandomForestClassifier utilizing early behavioral signals such as showroom visits, test drives, brochure downloads, and pipeline duration.
- **Results:** Significantly outperformed the majority-class baseline accuracy, achieving high ROC-AUC scores. Behavioral indicators such as test-drive completion and finance quote requests proved to be powerful predictors of serious buyers.

Model C: Vehicle Model Purchased (Multiclass Classification)

- **Target & Population:** model_purchased (Volt City, Volt Sedan, Terra SUV, Aura Premium) across converted buyers.
- **Algorithm:** Multiclass RandomForestClassifier evaluated using weighted-average Precision, Recall, and F1 scores.
- **Results:** Identified model confusion patterns between adjacent trims (e.g., sedan vs. city variants), providing actionable intelligence for showroom inventory and stock planning.

4. Service-Side Business Analysis

- **Service Costs & Centres:** Analysis of joined sales and service data via vehicle_id highlighted cost variations across service centers in Dubai, Abu Dhabi, and Sharjah.
- **Battery Health Degradation:** Scatter plot analysis mapping battery_health_pct against vehicle_age_months and odometer_km quantified long-term EV battery retention.
- **Workshop Efficiency:** Evaluated workshop wait days and customer satisfaction ratings, flagging repeat visits within 30 days to optimize after-sales support.

5. Strategic Management Recommendations

Based on empirical model outputs and data analysis, management should implement the following five recommendations:

1. **Prioritize High-Intent Leads:** Allocate senior sales personnel immediately to leads who request test drives and finance quotes, as Model B confirms these behaviors strongly correlate with conversion.

2. **Dynamic Inventory Allocation:** Use Model C's confusion patterns to adjust showroom stock levels in Dubai and Abu Dhabi, ensuring high-demand SUV and sedan models are readily available.
3. **Transparent Pricing Guidelines:** Leverage Model A's regression insights to standardize discount limits during deal negotiations, protecting profit margins across premium trims.
4. **Proactive Maintenance Scheduling:** Use service-center battery degradation trends to alert customers before significant range anxiety or warranty claims occur.
5. **Streamline Workshop Wait Times:** Address bottlenecks in service centres experiencing elevated repeat-visit rates to improve post-sale customer satisfaction scores.

6. Assumptions & Limitations

- **Correlation vs. Causation:** Predictive features (such as showroom visits) indicate strong association with purchasing behavior rather than direct mechanical causation.
- **Economic Variables:** Macroeconomic factors like UAE fuel prices or competitor EV pricing shifts were outside the scope of current internal logs.